

Eol Template Outline (+/- 1,500-2,500 words per Eol) – Due date CoB September 19

Please submit Eols to Masud Rana (MASUD.RANA@cgiar.org)

Section 0 – Title and contact Information

Title: Scaling Potato zero tillage and mulching in India

Flagship: Scaling Seed and Soil Fertility Management Systems for Resilient, Diverse, and Nutrition-Smart Farming

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Alignment: Sustainable Farming / Gender Accelerator / Digital Transformation

Section 1 – Innovation Package, Value Proposition & Alignment

The problem to be solved. Bihar, India, is among the country's poorest and most densely populated states, where rapid population growth (increase of 15% to 148 million people by 2036) intensifies pressure on food and nutrition systems. To meet future demands, more food needs to be produced in a sustainable manner. Agriculture, and particularly potato, is the backbone of livelihoods, yet potato-growing areas face degrading soils, falling groundwater tables, and residue burning after rice harvest that worsens air quality and greenhouse gas emissions. Farmers – who are primary beneficiaries of this project – also grapple with erratic rains, late-season heat waves, increased pest and disease pressure, and rising input costs—climate and economic stresses that lock many into low productivity and high risk. Women, who contribute substantially to farm labor, are disproportionately affected: they shoulder heavy drudgery in planting, weeding, and harvesting, yet have little say in farming decisions or access to technologies and training. Social norms and unequal resource access mean women farmers often work harder but earn less.

Your scaling ambition. The Indo-Gangetic Plains, which is home to about 6-7 million potato farming households, growing potatoes on +2Mha and producing +50Mt. Bihar alone has +800k potato farming households, growing potatoes on an average of 0.4ha of land. **The scaling ambition of this Scaling Solution Track is to transform conventional potato cultivation along the entire Indo-Gangetic Plains into conservation agriculture**, while starting with Bihar and focus on non-traditional potato cultivation in coastal areas where high salinity impeded agricultural activities. We expect to contribute most to supporting smallholder farmers to access agrifood solutions (50k households) and gaining better diets (+250k people), with a strong focus on reaching women (~50%), and transforming substantial agricultural land into regenerative agriculture (+20k ha). Beyond reach, we scale deep and expect (i) higher and more stable yields; (ii) ~25–30% lower production costs; (iii) net income gains; (iv) more jobs (particularly for women) and reduced drudgery of labour; (v) improved decision-making by women; (vi) elimination of crop residue burning and smaller CO₂-e footprint; (vii) improved soil health. Consumers benefit via

greater off-season supply. Public sector benefits include reduced residue burning, improved soil cover, and mainstreamable climate-smart extension.

Innovation & Value Proposition. PZTM is a validated bundle: place seed tubers on minimal/no tillage post-rice soil and mulch with rice straw or maize residue, enabling immediate planting, moisture conservation, weed suppression, easier harvest, and lower costs. PZTM is climate smart as the “cooling effect” of the mulch allows for later harvesting and thus planting by ~15 days; it also suppresses salinity in the soil as potatoes grow in the mulch making it an ideal innovation for coastal regions. The bundle is packaged with digital extension delivery model of proven community-videos, seed system development and mechanization. IPSR classifies the core practice at SRL ≥ 7 (demonstrated under real farm conditions); the limiting factors are complementary innovations: (1) community straw access/alternatives, (2) context-fit mechanization options for scale, (3) seed/market linkages, and (4) policy mainstreaming. Value proposition: PZTM reduces production costs while increasing yields resulting in higher profitability, while reducing drudgery and increasing soil health. For government and donors: a cost-effective, evidence-backed pathway to intensify rice fallows, curb burning, improve rural incomes and diets, and embed climate-smart extension. There are linkages with Sustainable Farming Science Program, Digital Transformation and the Gender Accelerator.

Customers & Alignment. End-users: smallholder growers (with a focus on women), plus new farmers adding potato in rice-fallow or maize-fallow niches. Institutional customers: the governments (Departments of Agriculture & Horticulture, district extension/KVKs), the research (ICAR-CIWA, ICAR-CSSRI, BARI), NGOs (JEEViKA, Nature Environment & Wildlife Society, ProdiPan, Women self-help groups), the private sector (Sheet Profile machinery manufacture, PepsiCo, McCain, Farmer Producer Organizations). The track aligns with Flagship 1.2 on *Scaling Smart Irrigation and Climate-Resilient Water Systems* and Flagship 2.1 on *Scaling Seed and Fertilizer Soil Fertility Management Systems for Resilient, Diverse, and Nutrition-Smart Farming*.

Evidence of Impact. On-farm pilots and trials in Bihar, coastal West Bengal and Khulna consistently show PZTM can halve irrigation events—doubling water productivity, advance planting by ~10–15 days, and raise marketable yield. Cost of cultivation drops roughly a quarter to a third by removing ploughing/ridging and reducing weeding; labor days fall substantially, with large drudgery reductions for women at weeding and harvest. Net margins and ROI improve even when yields are comparable, because cost savings dominate. Farmer field-days and side-by-side demos triggered organic diffusion; DG video analytics recorded strong reach and recall; JEEViKA districts saw rapid year-on-year expansion where SHG facilitators were active. Adoption rates are currently at ~5k while ongoing M&E activities could change this figure upwards. There is strong evidence of positive effects of our intervention on women equity, agency, and decision-making. Also, video-mediated extension effectiveness has been documented. For all sources, see the Reference list.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

Scaling Pathway & Sustainability.

Scaling pathway. The public extension program under the Department of Agriculture will be the main scaling pathway disseminating and scaling PZTM. Inclusion and mainstreaming of PZTM manuals into extension programs will ensure knowledge retention and institutionalization. A

secondary public scaling pathway will leverage NGOs such as JEEViKA, NEWS and Women self-help groups, and the private sector such as PepsiCo and McCain.

Delivery model. We blend three complementary channels. **(1) Digital extension:** video-mediated learning led by Digital Green, with short modules screened in community public viewings facilitated by women's Self-Help Groups (SHGs) and followed by group discussion. **(2) Traditional extension:** government extension agents and KVKs reinforce adoption through training calendars, seasonal advisories, and **side-by-side demonstration plots** that allow farmers to compare PZTM with conventional practices under local conditions. **(3) AI-enabled extension:** farmers and extension staff will have access to **Farmer.Chat**, an AI-based advisory platform providing timely agronomic guidance, troubleshooting, and customized information to complement human facilitation. A gender-responsive cascading training-of-trainers (ToT) model will ensure state, district, and block-level staff (30% female staff), JEEViKA Village Resource Persons (VRPs), and lead farmers (50% women) are skilled to coordinate these three channels effectively.

Sustainability and cost recovery. The track anchors delivery in existing public and private systems—so recurring costs are low and increasingly covered by routine budgets. Because PZTM is a knowledge-intensive, input-light practice, it is well suited to public dissemination at minimal marginal cost. To complement public finance, we will pilot cost-recovery instruments: **(i)** monetize verified **CO₂e reductions** (from avoided residue burning, fewer tillage passes, and improved soil cover) via voluntary carbon markets; and **(ii)** introduce a **PZTM-certified** scheme that enables buyers/CSR programs to contribute small premiums or service fees linked to climate and gender outcomes. Revenues will contribute to support public dissemination and training efforts.

Partnerships & Leverage.

CIP. A team of agronomic and socio-economic experts at CIP will provide technical leadership (agronomy, adaptation, mechanization), lead MELIA (adoption, yield/costs, GESI indicators) and learning studies on testing effectiveness of delivery mechanisms; steward Scaling Readiness and Scaling Scan updates; facilitate multi-stakeholder engagement and act as international focal point.

IRRI. A team of agronomic, socio-economic and engineering experts will identify potential scaling regions by analyzing crop rotation timing and rice varieties. They scale the innovation by their networks with rice growers.

Partnerships and evidence of demand are as follows:

Departments of Agriculture & Horticulture. In Bihar, Chair of the State PZTM Coordination Committee; integrate PZTM into regional extension programs; mandate/implement block-level training and monitoring. PZTM as a climate smart agricultural system aligns fully with the State's Agricultural Road Map (2023-2028). There is strong demand from the government of Bihar for scaling PZTM. In the context of a bilateral project that started in 2022, CIP has built strong relationship with the DoH and DoA of Bihar. CIP was invited to submit a "Scaling Proposal for PZTM in Bihar" which is now being discussed at the DoA with State-level funding coming from 2026 onwards, if approved. At the same time, discussions take place for PZTM manuals to be included in the extension programs and routine training. The states of Odisha and West Bengal

support CIP for the establishment of potato seed system through which a mechanism to mainstream PZTM in both seed and food production has been established.

Digital Green will be responsible for the technical parts revolving around the digital extension. In terms of the extension videos, DG will create/update 6–8 video modules on various aspects of PZTM (site selection, residue handling, seeding, mulching, irrigation/fertilizer tweaks, pest/rodent management, harvest & straw reincorporation). They will also spearhead the refinement of PZTM knowledge in DG's Farmer.Chat which is already trained on PZTM manual. They will also train JEEViKA staff and leverage their networks of village resource persons to conduct monitoring and evaluation activities. CIP has an existing agreement with DG.

JEEViKA provides a vast network of women self-help groups and village resource persons. These have already been leverage during the initial pilot and roll-out of PZTM in several districts in Bihar. They will host screenings/demos, identify and mentor 1,000+ women promoters; facilitate women's access to seed and small credit; channel feedback on social impacts.

Gesellschaft fuer Internationale Zusammenarbeit (GIZ) has be a strong interest in scaling PZTM across India and Sout-east Asia in more general. PZTM is included as one of the flagship innovations in the GIZ i4Ag portfolio. Leveraging GIZ excellent contacts and networks across India, they will help broker policy dialogues; they will help support the development of linking PZTM to voluntary carbon markets and creating/promoting mechanization prototypes. Jointly with GIZ consultants of its International Business Unit, we have pitched PZTM to private sector in India (e.g., PepsiCo).

Leverage. There are several CGIAR-external funding and co-investment opportunities: (1) the ongoing GIZ bilateral project; (2) the Bihar Department of Agriculture has signaled a scaling project from 2026, (3) private sector (e.g., PepsiCo, McCain) have been investing in conducting agronomic validation of PZTM prior to large-scale engagement in 2026 and a machinery manufacture Sheet Profile in collaboration with Odisha University of Agriculture and Technology invests in developing a prototype of mechanized PZTM; (4) World Bank agriculture operations are an active avenue—recent discussions explored embedding PZTM modules, (5) In-kind contribution from Nature Environment and Wild Society to promote PZTM in coastal regions in West Bengal from 2025. S4I funds act as catalytic capital to facilitate responsible and cost-effective scaling.

Risks & Bottlenecks.

Straw availability & logistics. Competing uses (fodder), moisture at rice harvest, and transport bulk can constrain mulching at scale. Mitigation: promote synchronized mulching in clusters; pilot straw banks and paid straw services (youth micro-enterprises); test alternate residues where rice straw is limited; link with residue-management/CSA schemes for balers/transport vouchers.

Mechanization of rice harvesting. Increasing adoption of mechanization of rice harvesting makes it difficult for farmers to have sufficient rice straws and four-wheel tires can create grooves in the plots, IRRI and CIP with an agricultural machinery manufacture have been developing the prototype of mechanized PZTM to address this issue.

Mechanization of potato planting and harvesting. The labour-saving advantage of PZTM is based on manual potato cultivation. In the mechanized conventional potato cultivation, PZTM is more labor intensive and has less incentive. Our target is smallholder farmers with manual potato production, meanwhile mechanized PZTM will be developed to expand PZTM to large-scale mechanized potato production areas.

Seed availability. Coastal regions are far from potato seed supply chains. Mitigation: collaborate with bilateral projects of seed system development and facilitate the private sector involvement in production and delivery of salt-tolerant potato varieties.

Rodents/stray cattle & tuber exposure. Mulch can attract rodents; open fields risk cattle disturbance. Mitigation: community rodent campaigns (trapping, field sanitation), plot-edge barriers where feasible, synchronized planting to dilute risk; include practical modules in videos; monitor hotspot blocks and respond quickly.

Market/storage pressure. Localized production spikes could stress cold storage/marketing. Mitigation: phase area increases; link clusters to buyers and storages; promote on-farm low-cost storage; use market intelligence to stagger harvests.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

Readiness & Responsible Scaling¹.

IPSR places **PZTM at SRL ≥7**; two important enablers, such as “straw access” (readiness score – low) and “mechanization” (readiness score – moderate) were identified as key bottlenecks at scale. While these constraints are non-trivial, we envision the availability of straw mulch and mechanization (to scale to more labour-constraint and industrialized State’s) not to be a key challenge during the current scaling phase. The *Scaling Scan* rated technology and knowledge strong; finance, value-chain, public-sector governance are moderate and must be intentionally strengthened. *GenderUp* surfaced both opportunities (women’s labor relief, leadership via SHGs) and risks (men’s appropriation of a successful enterprise; landless labor displacement). The project therefore (i) commits to couple-sensitization and targets women for dissemination to solidify agency, (ii) tracks labor and inclusion effects (disadvantaged groups, caste, youth), and (iii) institutionalizes PZTM so benefits persist beyond project life. Environmental safeguards are intrinsic (no burning, residue to soil), with agronomic guardrails (rodent control, IPM under mulch) embedded in content and ToTs.

Achievable target, Resources & Activities.

Target. 50,000+ farmers adopting PZTM statewide by end-program, with ≥50% women participants, ~20,000 ha under PZTM/ conservation agricultural practices, and 1,000+ women in promoter/lead roles. Outcomes: substantial on-farm cost reductions, increase in rural incomes,

¹ This sub-section’s results are based on utilizing a LLM trained on PZTM knowledge, evidence, and methods, as well as IPSR, Scaling Scan, and GenderUP techniques. The results have been verified by a human for adequacy and correctness.

diet diversification, increased household resilience, women empowerment and better jobs, and measurable climate co-benefits (e.g., residue retention, no burning).

Resources. Staff equipped with technical knowledge of PZTM and scaling processes are required. On-the-ground existing staff will be involved to draw on existing relationships with key partners, such as Digital Green, DoA, and DoH. Where expertise is not available, consultancies are optional. Strong institutional standing and relationships with Bihar government, and ICAR exist and will be leveraged.

Activities.

Month 0-6. Scaling Strategy: a multi-stakeholder gathering including all relevant stakeholders, such as DoA/DoH of Bihar, private sector, Digital Green, JEEViKA, regional development banks, GIZ and other donors, ICAR institutes, etc., will be organized to develop and approve a scaling strategy.

Month 3-9. Institutional anchoring: PZTM inclusion in Bihar extension program and development of training materials.

Month 3-9. Update 6–8 video modules (including straw logistics, rodent control, fertilizer/irrigation tweaks); finetune PZTM content for LLM built into Farmer.Chat.

Month 10-14. Training extension officers, JEEViKA staff on PZTM and implement community screenings; SHG-hosted demos (side-by-side) in every target block; advisory through Farmer.Chat.

Month 14-20. Analyzing delivery mechanisms (i.e., video-mediated extension vs. AI-based extension) and examining collected M&E data.

Month 22. Mid-project course correction meeting with stakeholders to agree on current scaling path or if adjustments are needed with specific focus on responsible and sustainable scaling.

Month 23-36. Further scaling to additional Districts in Bihar.

Timeline, Milestones & Monitoring.

Year 1 (2026): Anchor policy (state advisory, inclusion in plans); Scaling Strategy approved; MoU with DoA/DoH signed; trained some 50+ extension officers and 50% JEEViKA staff in priority Districts; extension videos updated; Farmer.Chat LLM content finetuned; reach 10,000 adopters; MELIA and GESI baseline.

Year 2 (2027): Expand to remaining potato districts; institutionalize refresher extension officers; reach 30,000 adopters; interim MELIA, GESI data to inform course correction and updated Scaling Strategy.

Year 3 (2028): Statewide coverage; govt budgets carry majority of demo/training O&M; custom planting services active in ≥8 districts; reach 50,000+ adopters; publish MELIA synthesis; PZTM included in state extension manuals; exit plan enacted with DoA/DoH and JEEViKA.

Monitoring. A joint MELIA system (DoA/DoH–JEEViKA–DG–CIP) tracks: adoption (by gender/caste), area (ha), yield/costs/irrigations, service roles created, consumer reach proxies (cluster output), and institutional uptake (budget lines, advisories). Short-loop learning updates videos/micro-plans mid-season; long-loop reviews adjust strategy annually. Responsible-scaling check-ins (GenderUp) are held pre- and post-season to surface and address unintended effects.

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Annexes

Appendix A – Scaling Scan Analysis

- Technology/Practice: PZTM (Potato Zero Tillage and Mulching) – Strengths:** Proven agronomic benefits in Bihar; on-farm trials show ~14% higher potato yields and 24% higher marketable yield versus conventional tillage. The practice saves 10–15 days in planting time (plant right after rice harvest) and cuts irrigation water use by ~50%, doubling water productivity. Farmers eliminate plowing and reduce weeding/harvesting labor, saving up to 24% in production costs. This boosts profit margins (one survey found PZTM net profit ₹18,900/ha vs ₹2,730/ha under traditional methods, ROI 21.7% vs 2.4%). GHG emissions per hectare drop ~42% due to less fuel use and no residue burning. **Weaknesses:** Some agronomic challenges persist. Mulching can attract rodent pests, requiring new management (farmers have reported rats damaging up to 25% of tubers under straw if not controlled). Farmers must secure sufficient rice straw; **mulch availability** is a key constraint, as many pilot farmers cited difficulty obtaining enough straw. In areas where rice residues are used for fodder or harvested by combine, mulch material may be limited. Additionally, if rice harvest is delayed or fields waterlogged, it can still constrain potato planting time (timing conflicts with preceding crops were noted). Scaling will require addressing these technical gaps (e.g. community rodent control, alternate mulch sources) to maintain PZTM's performance at larger scale.
- Awareness & Demand: Strengths:** The pilot demonstrations (2021–2023) generated significant local awareness and interest. In 120 villages across Nalanda, Patna, Rohtas, Vaishali, over 460 farmers were trained in the first year. Hands-on demos (often in women's kitchen gardens) and side-by-side comparisons with conventional plots built farmer confidence – neighbors observed faster crop emergence, healthier plants, and cleaner, larger tubers under straw mulch. Many initially skeptical farmers (including men who doubted potatoes could grow without soil) became convinced after seeing a successful PZTM harvest. Government agencies are also increasingly aware – the practice aligns with Bihar's climate-smart agriculture agenda, sparking interest from the Department of Agriculture/Horticulture. **Weaknesses:** Beyond these pilot communities, overall awareness in Bihar's farming population remains low. PZTM is a novel concept, and demand is nascent – most of the ~800,000 potato growers in Bihar have not yet seen or tried it. Scaling to >50,000 farmers will require an intensive campaign of communication and demonstration. Social norms can be a hurdle; early women adopters had to persuade their families to allow trying PZTM. Until tangible success is witnessed, many farmers (and their spouses) are reluctant to deviate from conventional practices. Thus, creating widespread demand will hinge on peer learning and visible success stories. The project must invest in **mass awareness channels** – e.g. village demonstrations, video extension (Digital Green), and local champions – to convert skepticism into curiosity and demand.
- Business Cases: Strengths:** PZTM offers a compelling economic proposition for farmers. By slashing land preparation and input costs, it raises profit per hectare substantially (farmers' gross margins improved by ~₹15-20k/ha in pilots). The return on investment can be several times higher than conventional practice, especially in stress-prone areas. This means smallholders have a clear financial incentive to adopt – **more crop for less cost**. Additionally, PZTM opens new business opportunities along the value chain. For instance, potato processors (e.g. PepsiCo, McCain) have shown interest in the

technique: CIP facilitated visits for these companies to assess whether zero-till, mulched potatoes meet their quality needs and could serve as seed or processing supply. If corporate buyers recognize advantages (e.g. more uniform tubers, earlier harvest), it could create contract farming or premium market channels for PZTM adopters. Mechanization service providers also see a business case emerging – custom operators could offer zero-till potato planting or straw mulching services, tapping into demand as area expands.

Weaknesses: Some elements of the business model are still unformed. There is not yet a well-developed market for **rice straw as a commodity** – farmers currently rely on their own field's residues, and surplus straw trade or alternative mulches are not common practice. This could limit expansion in areas where straw is scarce or has competing uses (e.g. livestock feed). Also, input suppliers and machinery companies have only begun to engage; specialized planters or mulch spreaders are not widely available in Bihar yet, meaning entrepreneurial service provision is in its infancy. The value chain for potatoes itself could present risks: if PZTM dramatically increases production in a locality, farmers might face glut or storage issues unless market linkages improve. Recognizing this, project partners highlighted the need for **strengthened storage and market linkages** as scaling enablers. Ensuring a robust business case at scale may involve integrating PZTM into procurement schemes (e.g. processors sourcing low-carbon potatoes) or leveraging carbon credits (PZTM abates ~720 kg CO₂/ha by avoiding residue burning, an added value that could be monetized via climate finance). These business model elements will need cultivation alongside farmer adoption.

- Value Chain Integration: Strengths:** PZTM fits relatively seamlessly into the existing potato value chain in Bihar, with added benefits. The end product (potatoes) is the same crop and variety farmers already grow, so market acceptance is high – in fact, zero-till potatoes often have *attractive skin color and brightness*, fetching a good market price. Early planting under PZTM can bring potatoes to market ~2 weeks sooner, potentially capturing better prices or avoiding late-season heat damage. The technique also enables potato cultivation in marginal areas (like sodic soils or high-residue fields) that were previously uncultivable, which could expand overall production and involve new producers. There are promising signs of value-chain actor engagement: as noted, multinational processors have been briefed on PZTM, and the government sees it aligning with state efforts (e.g. promoting zero-till wheat, encouraging residue use). **Weaknesses:** Certain value-chain bottlenecks could impede scaling if unaddressed. One is **quality seed supply** – many smallholders use saved potato seed or suboptimal varieties. PZTM can boost yields, but if farmers lack good seed, the full benefit won't materialize. The project has identified this and is exploring partnerships to ensure quality seed availability (including possibly using PZTM harvests as seed source if they prove healthy). Another challenge is post-harvest handling. Because PZTM tubers form under straw (not soil), they tend to be very clean and undamaged, which is good for market, but farmers must adjust harvesting and storage practices (no heavy digging required, but need to collect tubers from straw and protect them from rodents or weather during drying). Cold storage access remains a general issue in Bihar – a surge in production from PZTM areas might strain local storage if not planned for. Lastly, if traditional supply chain middlemen do not understand PZTM, there could be misconceptions (though the product is essentially the same potato). Continuous engagement with value-chain actors (traders, wholesalers, processors) is needed so that increased output and potentially different harvest timing translate into smooth market absorption rather than oversupply or farmer price penalties.

Integrating PZTM into contract farming or FPO (farmer producer organization) channels could be a way to secure markets for growers at scale.

- Finance: Strengths:** At the farm level, PZTM reduces the need for external financing because it lowers production costs. Farmers save on expenses like tractor hire, diesel, and agro-chemicals (a 25–32% cost reduction per hectare), which means less out-of-pocket cash or credit is needed to grow the crop. This makes potato cultivation more financially accessible to resource-poor farmers, potentially reducing their debt burden. The practice is thus financially *self-reinforcing* – once a farmer adopts and sees higher net income, they can reinvest profits rather than borrowing for the next season. For scaling partners, initial funding is secured via a GIZ-supported project (Oct 2021–Dec 2025) and co-investment from CIP. Looking forward, the Government of Bihar’s involvement provides a path for sustained financing: integration into state schemes can unlock public funds (e.g. using national programs like RKVY to fund trainings or subsidize equipment for PZTM villages). **Weaknesses:** Some financing gaps must be managed. While implementing PZTM is cost-saving overall, farmers might face *upfront needs* – e.g. acquiring sufficient straw or transporting it from elsewhere, which could incur costs if not readily on-farm. If mechanized planters or balers are introduced, farmers or service providers will need capital or credit to purchase them. Without subsidies or credit lines, this could slow adoption of equipment. The current scaling initiative relies on donor funding and government budget allocations; any discontinuity in these could hinder momentum. Ensuring financial sustainability means institutionalizing PZTM into regular extension budgets and possibly private sector CSR investments (given its climate benefits). Another consideration is the **opportunity cost** for laborers: PZTM saves labor (male labor days drop ~33%), which is good for farmer finances but could reduce wage opportunities for landless laborers (a social equity concern addressed in the Gender section). A just scaling approach might explore alternative livelihoods or engagement (such as employing local youth as PZTM service providers or straw aggregators) so that labor savings don’t translate into net income loss for the rural poor. Overall, the financial outlook for PZTM is positive, but creative financing mechanisms (microcredit for inputs, equipment subsidies, outcome-based incentives for climate-smart practices) may enhance and accelerate scaling.
- Knowledge & Skills: Strengths:** A substantial knowledge base has been built around PZTM, and capacity development is a core part of the scaling plan. CIP has documented the practice in an agronomic manual and has years of trial data to inform recommendations. Training materials are available in user-friendly formats – e.g. six short training videos in local language featuring farmer trainers were produced and screened in communities. Digital Green brings expertise in developing and disseminating these educational videos. Frontline staff (government extension workers and JEEViKA village resource persons) have already been trained on PZTM techniques during the pilot. The approach is relatively simple to learn (no complex machinery for basic implementation), and many farmers grasped it through one demo season alongside their normal practice. Ongoing agronomic support is planned: extension agents and lead farmers will be coached on season-long management, troubleshooting (e.g. what to do if mulch blows away or how to adjust fertilizer in zero-till), ensuring that knowledge is deepened over time. **Weaknesses:** Scaling to tens of thousands of farmers poses a major training challenge. Currently, knowledge about PZTM is concentrated in pilot districts and among a small cadre of trained personnel. Extension services in Bihar are stretched thin; Digital

Green noted the difficulty of monitoring activities in 120 pilot villages without dedicated staff. To reach scale, significant **capacity building** at multiple levels is required – from state agronomy experts down to community facilitators. The project must avoid “knowledge dilution” – i.e. incomplete or incorrect understanding leading to partial adoption (for example, if someone skips the mulching step, results suffer). This means training needs to be high-quality and repeated. Farmer-to-farmer learning will be crucial but can only happen if early adopters are well supported and documented. Another gap is the integration of new knowledge into formal channels: PZTM isn’t yet in standard training curricula or ag extension manuals (one of the objectives is to include it). Until it is, extension agents may lack reference materials or incentives to promote it. Moreover, certain skills need reinforcing – e.g. pest surveillance under mulch (which differs from conventional practice) or proper straw handling. Continuous learning mechanisms (feedback loops, field days, radio/WhatsApp advisories) will be needed so that farmers and extensionists can share experiences and solutions. In summary, the knowledge exists, but scaling it up geographically and ensuring consistent quality of skills and advice is a task requiring dedicated resources.

- Collaboration: Strengths:** Multi-stakeholder collaboration is a cornerstone of this scaling effort, and roles have been clearly delineated to harness each partner’s strengths. The Government’s Department of Agriculture (and Horticulture) will provide institutional leadership – chairing a State PZTM Coordination Committee and aligning the initiative with government programs. Digital Green (DG), with its specialization in video-mediated extension, will lead digital content creation and training of mediators. JEEViKA (Bihar’s rural livelihoods mission) leverages its extensive network of women’s self-help groups for community mobilization and last-mile delivery. CIP contributes technical expertise, ongoing R&D, and backstopping support (with potential input from national research institutions like ICAR’s Central Potato Research Station). This consortium is further bolstered by the GIZ project platform, which provides funding and a scaling lens (the GIZ team brings scaling specialists and cross-country learning). Such collaboration has already proven effective in the pilot: CIP, DG, and JEEViKA coordinated to train farmers through a mix of digital and group-based methods, and government extension workers participated, lending credibility and reach. **Weaknesses:** With many partners comes the challenge of coordination and communication. The pilot revealed that without dedicated field coordination, some activities (e.g. ensuring all planned video screenings happened) slipped through the cracks. To scale successfully, robust coordination mechanisms are needed – the plan includes regular multi-partner review meetings and dedicated project staff to liaise between organizations. Still, potential overlaps or gaps must be managed (for example, both JEEViKA and Ag Dept extension have field personnel; they will need clear division of labor to avoid confusion at village level). Ensuring all collaborators share data and learnings in real time is another hurdle – a joint M&E system is proposed to address this. Additionally, private sector involvement (beyond GIZ) is nascent; getting companies like PepsiCo on board as true partners (e.g. providing buying contracts or extension support) is still in exploratory stages. If successful, that could add complexity but also resources. Lastly, long-term collaboration needs incentives: after project funding ends, partners like DG or CIP will need continued support or institutionalization to remain engaged. Formalizing roles (e.g. MoUs with the state to keep using DG’s videos, or CIP advising a state technical committee) could help sustain collaboration. In summary, the partnership is strong and quite inclusive (public, NGO, research, donor, community

organizations), but it will require active management and flexibility to respond to on-ground realities during scaling.

- Evidence & Learning: Strengths:** The PZTM scaling initiative is grounded in robust evidence and has a strong learning orientation. Over 2+ years, CIP and partners have generated a **rigorous evidence base**: multiple on-station and on-farm trials in Bihar and similar contexts, plus socio-economic studies. Notably, a randomized controlled trial across 200 villages (2,000 households in Bihar) was conducted to quantify PZTM impacts. Findings consistently show yield increases (~15% in India on average) and large gains in saline areas (up to 3-4x yield in Bangladesh), as well as input cost reductions ~30% and major ROI boosts. These results have been documented in scientific publications and briefs (e.g. Ramírez et al. 2022/2024, Kakraliya et al. 2024). The program tracks not just agronomics but also gender and climate metrics – a dedicated Gender research study (Digital Green 2024) evaluated women’s empowerment outcomes, and soil health and GHG data are being collected (550 soil samples analyzed to assess organic carbon changes, etc.). This evidence is being actively used to inform scaling strategy and convince stakeholders: a summary of PZTM benefits and a draft scientific manuscript were submitted to the Dept. of Horticulture as part of a proposal to mainstream PZTM. On learning mechanisms, the project has a strong MELIA plan. They will use tools like Kobo Toolbox for field data collection and feedback loops. During rollout, extension officers and JEEViKA staff will continuously capture farmer feedback (e.g. challenges encountered) and the coordination team will document and respond with adjustments (for instance, if rodent issues spike, they’ll record it and introduce mitigation measures promptly). The initiative is treating scaling as an iterative learning process – e.g. updating training content based on field experience and producing new videos for emerging issues in later years.

Weaknesses: One potential gap is **real-time data flow** from the field. The pilot phase noted challenges in fully monitoring activities across sites. For scale-up, Digital Green and CIP plan dedicated M&E personnel, but ensuring timely collection and analysis of adoption rates, yields, and gender outcomes will be complex. There is also the risk that learning remains within the project team – a wider community of practice (including state officials, other CGIAR initiatives, etc.) should be looped in so that lessons influence broader scaling efforts. Another consideration is the **evidence perception** among farmers: while scientific evidence is strong, ultimately many farmers will be convinced by seeing local results and hearing from peers more than by statistics. Thus, embedding farmer-led assessments (farmer field days, local “evidence demos”) is as important as formal trials. Lastly, as scaling progresses, new challenges may arise that weren’t seen in small pilots (e.g. if a certain district has a unique pest). The program’s success will depend on its ability to detect and learn from these surprises quickly. The planned adaptive management approach – with systematic problem tracking and solution dissemination – is intended to address this, but it will need diligent execution. Overall, the culture of evidence and learning in this project is strong, reducing the chances of blindly pushing an innovation without feedback, and increasing the chances of long-term success.
- Leadership & Management: Strengths:** There is high-level buy-in and clear leadership structures for scaling PZTM in Bihar. The Bihar Department of Agriculture (and allied Dept. of Horticulture) is poised to take ownership – they will chair the State PZTM Coordination Committee, ensuring the initiative is anchored in government priorities. This leadership is key to integrating PZTM into official programs (e.g. they can issue advisories to District officers to promote it, include it in state agricultural roadmaps). The project management

plan delineates responsibilities: government extension officers serve as frontline managers for field implementation and data reporting, while CIP provides technical guidance and oversight. Digital Green and JEEViKA manage their components but under the umbrella of the state-led plan, with regular joint reviews. Such a structure should foster accountability and alignment. Already, the Government of Bihar has signaled leadership by identifying zero-till residue management as a climate-smart priority and looking to institutionalize PZTM post-project. On the project side, CIP has dedicated staff coordinating day-to-day activities and liaising with government counterparts. GIZ's involvement has brought in scaling management expertise and support for strategic planning (including employing the CGIAR Scaling Scan and Scaling Readiness tools to shape the proposal). **Weaknesses:** Effective management will require maintaining momentum and coordination through political or administrative changes. Government leadership is an asset, but also a risk if key champions are transferred or if other priorities take precedence. To mitigate this, the program aims to cement PZTM into institutional processes by Year 5 (e.g. making it part of routine extension mandate) – however, until that happens, active advocacy and relationship-building with bureaucratic leadership is needed. Another management challenge is multi-level oversight: activities will span state, district, block, and village levels. Ensuring good communication and reporting up and down this chain is non-trivial. The project's plan for quarterly or biannual multi-partner review meetings is good, but day-to-day issue resolution will fall to the coordination team – they must be empowered to make quick decisions (e.g. reallocating resources if one district lags behind). The leadership will also be tested in managing incentives and recognition among staff: field officers and community volunteers need encouragement to promote a new practice that may not yet be in their formal job description. The proposal suggests modest incentives or recognition for those who excel (e.g. awards for SHGs that champion PZTM), which could help maintain enthusiasm. Lastly, strategic management of scale is crucial: the plan is to start in high-potato districts and gradually expand. Sticking to this phased strategy (and not expanding too quickly without support) will require discipline from leadership. In essence, strong leadership commitment exists, but proactive management and adaptability will be needed to navigate the complexities of scaling an innovation through large bureaucratic and community systems.

- **Public Sector Governance: Strengths:** The enabling environment in Bihar is favorable for PZTM scaling. The practice aligns closely with public policy goals – improving smallholder incomes, sustainable intensification, climate resilience, and women's empowerment are all highlighted in Bihar's agri-development plans. PZTM directly supports these by raising productivity without extra land, recycling crop residues (reducing pollution), and easing women's labor burdens. Recognizing this, the Departments of Agriculture and Horticulture are supportive; they have been engaged with evidence sharing and discussions to include PZTM in extension programming. A key governance advantage is the existing platforms the state can leverage: for instance, the government's **ATMA** (Ag Technology Management Agency) extension network down to blocks, and JEEViKA's network of SHGs (a government-supported program), which together form a channel to reach farmers in an organized way. Integration with national schemes is also on the table – e.g. climate-smart agriculture funds or machinery subsidies can be directed to promote components of PZTM. By embedding PZTM in such schemes, the public sector can institutionalize support (financial and advisory) for the long term. **Weaknesses:** Some governance-related challenges could affect scaling. Bureaucratic procedures for approving new demonstrations or subsidies might be slow, potentially delaying the rollout

pace. PZTM is not yet part of any *official package of practices* for potato – getting it formally endorsed in state agri advisories or KVK (Krishi Vigyan Kendra) recommendations will take effort. Until then, some local officials might be hesitant or uncertain about promoting it. Additionally, there could be competing priorities; for example, if the state pushes a different potato initiative (such as promoting another input or variety), extension agents might be diverted. The governance context also includes informal norms – some village leaders or local influencers may need to be convinced so they don't discourage farmers from “doing something different.” Another issue is equity in public support: the government must ensure that benefits of scaling (trainings, any subsidies like for straw or seed) reach marginal and remote farmers, not just well-connected ones. JEEViKA's involvement helps in targeting poorer households, but close monitoring of inclusion is required. Finally, sustainability of governance support is crucial – a change in government or donor priorities could shift focus. The mitigation for this is to **mainstream PZTM into routine programs** by 2030 (e.g. including it in the State Agricultural Roadmap and budgeting for it beyond the project). Achieving that will require continuous advocacy and demonstrating results at scale to policymakers. In summary, the public sector context is generally enabling, but formal institutionalization and equitable implementation are areas to be safeguarded as PZTM scaling progresses.

Appendix B - Innovation Packages & Scaling Readiness (IPSR) Analysis

Core Innovation: *Potato Zero Tillage and Mulching (PZTM) practice* – This is the central innovation being scaled. **Scaling Readiness Level (SRL):** PZTM is assessed at SRL ~7 (on a 9-level scale), indicating it has been proven in real-world settings and is ready for wider scaling. Since 2018, CIP and partners have developed and validated PZTM across multiple environments (on-station trials, on-farm pilots in Bihar, Assam, Bangladesh). By 2023, the practice showed consistent agronomic and economic benefits in Bihar (yield gains ~10–15%, cost savings ~25%, profit increases, etc.) and farmers have adopted it in pilot villages. A comprehensive manual and training videos exist, reflecting a mature technical package. **Use level:** Currently, usage is limited to pilot implementations (a few hundred farmers in 4 districts) and experimental trials. So while the innovation itself is ready, its uptake at scale in the target context is only nascent. The core innovation appears “**scalable**” in principle – it's low-cost, relatively simple, and aligns with local farming systems – but scaling readiness analysis highlights that certain *complementary innovations* and conditions must accompany PZTM for successful widespread adoption. Key bottlenecks lie not in the core practice's efficacy, but in supporting factors that need to catch up in readiness or usage.

Complementary Innovations & Bottlenecks: To effectively scale PZTM, a package of complementary innovations and enabling solutions must be in place. The main complementary components, their current readiness/use levels, and bottlenecks/strategies are:

1. Digital & Community Extension Model (Video-Mediated Learning via SHGs):
Innovation: Using Digital Green's video-based training and JEEViKA's community networks to disseminate PZTM knowledge. **Readiness:** High – This approach is well-established in Bihar for other practices, and for PZTM it was piloted with success (6 farmer-friendly videos were produced and shown in 120 villages). Extension staff and women's group facilitators (VRPs) have been trained to use battery-powered projectors and lead discussions. **Use:** Moderate – During the pilot, screenings reached a limited audience, and DG found it challenging to monitor all villages without dedicated field personnel. Beyond the pilot area, usage of this innovation is minimal so far. **Bottlenecks:**

Scaling this approach requires resources and coordination. There is a need for more **localized content** (additional videos in local dialects, covering emerging issues like rodent control) and a system to ensure regular screenings. In the pilot, lack of sufficient on-ground coordinators meant some communities didn't get all the planned sessions. *Scaling Strategies:* The project will fund **dedicated dissemination coordinators** through DG to support and track video sessions. A joint micro-planning with JEEViKA will align video showings with SHG meeting schedules. New content will be created iteratively in response to farmer feedback (e.g. if a knowledge gap or myth is identified, an instructional or testimonial video will be produced). By institutionalizing the digital approach within the extension system (e.g. making PZTM videos part of the Dept. of Agriculture's extension toolkit), this complementary innovation will become routine. Overall, ensuring the *extension delivery innovation* is fully staffed and scaled (rather than an afterthought) is critical to drive PZTM awareness and correct implementation. With these steps, the **readiness** of this extension model will translate into **wider use** (reaching thousands of groups), overcoming the current dissemination bottleneck.

2. Straw Supply and Alternate Mulching Solutions: *Innovation:* Systems or methods to ensure farmers have adequate mulching material (rice straw or alternatives) for PZTM at scale. This includes innovations like community straw banks, straw collection services, or using other organic mulches if rice straw is unavailable. *Readiness:* Low – Presently, there is no formal solution in place; the pilot relied on farmers using their own paddy straw. In some cases, mulch availability was a *binding constraint* – farmers voiced that without enough straw, they could only implement PZTM on small plots. Research into alternative mulches (e.g. other crop residues, chopped maize stalks, etc.) has been identified as a need but not yet done. *Use:* Low – Outside of those who had on-farm straw, farmers have not adopted any systematic way to secure mulch. There's also competition for straw (fodder, thatch, etc.) in rural communities. *Bottlenecks:* This is a critical bottleneck – if mulch cannot be sourced affordably, PZTM uptake will stall. Challenges include: transporting bulky straw to fields, opportunity cost of straw (farmers who keep livestock might be reluctant to mulch instead of feed), and a lack of equipment for efficient straw handling (most smallholders collect and spread straw manually). The pilot even observed incidents of mulch being disturbed by animals when left in fields, pointing to the need for community-wide solutions (one farm mulching alone can attract pests or free-grazing cattle). *Scaling Strategies:* The strategy has multiple prongs to address this. First, **optimize on-farm straw use:** by promoting synchronized rice harvesting and mulching among neighboring farms, so that straw is immediately used and fields aren't isolated (detering pests). Second, explore **alternative mulch materials** – for example, testing if sugarcane trash, wheat straw, or leaf litter could supplement rice straw. The Gender research recommends R&D on alternates to reduce dependence on rice straw alone. Third, enable a **straw supply chain:** encourage local entrepreneurs or cooperatives (perhaps youth or landless groups) to collect excess straw from those who don't need it and deliver it to PZTM adopters for a fee. The government can facilitate this by linking to residue management schemes (Bihar already discourages burning; it could support balers or transport for straw). Additionally, the program will educate farmers on post-harvest straw management – e.g. after potato harvest, incorporating remaining straw into soil (turning a potential waste into soil organic matter), which might incentivize farmers to value mulch. By improving the *readiness* of straw supply mechanisms (through innovation pilots like straw banks or subsidized balers) and promoting their *use* community-wide, this bottleneck can be alleviated. Success on this front is pivotal – the mantra is “zero tillage

and mulching”, so the scaling package must make mulch accessible; otherwise, partial adoption (zero-till without mulch) will give poor results.

3. Mechanization for PZTM (Zero-Till Potato Planter and Related Tools): *Innovation:*

Introducing or developing planting machinery suitable for zero-till, mulched potato cultivation. A specialized **zero-tillage potato planter** that can place seed tubers in untilled soil and possibly help spread or handle straw is considered a complementary innovation.

Readiness: Moderate – The concept is known and mechanization is seen as a scaling enabler, but solutions are at an early stage of adaptation to PZTM contexts. CIP has begun work on this front: a training module on “Potato Zero Tillage Mechanization (PZT-Mech)” was developed, indicating prototype machinery or techniques are being evaluated. In regions like Punjab, existing no-till planters (for other crops) or modified machines could be tried for potatoes. However, as of 2025, most PZTM in Bihar has been done manually or with very basic tools.

Use: Very low – No significant use of mechanized planters in the Bihar pilots; farmers and laborers placed seed tubers and mulch by hand in small plots. For larger-scale operations (or bigger farmers), manual implementation could be labor-intensive, so mechanization is not yet utilized but will be needed.

Bottlenecks: Without mechanization, scaling to tens of thousands of hectares may be slow or limited to small plots. Bottlenecks include: lack of locally available equipment designed for this practice, little awareness or technical capacity to use such machinery, and cost. Additionally, mulched fields present challenges for machines (straw can clog equipment).

Scaling Strategies: The project has identified **context-specific mechanization** as crucial.

Strategic steps include partnering with agri-engineering researchers and manufacturers to design or adapt planters for PZTM. For example, retrofitting a zero-till seed drill with a tuber metering device, or a planter that can cut through straw and soil to plant potatoes at proper depth. Pilot tests of such equipment will be conducted on research stations and lead farmers’ fields. The government can incentivize this by offering subsidies on any approved zero-till planters (under climate-smart mechanization schemes) to early adopters. Another strategy is to engage custom hiring centers – if machinery is too expensive for individual smallholders, local service providers can own it and rent out planting services. Training tractor operators and service entrepreneurs on PZTM machinery will go hand in hand. Over the five-year scaling program, the aim is to have at least a few mechanized options validated and in use in high-potato belt districts, to demonstrate feasibility. As the technology is refined, *readiness* will improve (moving from prototype to fully field-ready machine), and with subsidy and demonstration, *use* will increase (more farmers opting for machine planting, especially those with larger plots). The benefit will be faster planting on more area and reduced drudgery, reinforcing the scalability of PZTM beyond what manual labor alone could achieve.

4. Enabling Policies and Institutional Integration: *Innovation:*

Policy and institutional changes that embed PZTM into the mainstream agricultural extension and support system. This includes official recognition (e.g. inclusion in state agronomy advisories, government schemes providing inputs for PZTM, extension mandates to promote it) and incentive structures (like climate change programs rewarding zero-till adoption).

Readiness: Moderate – Government interest is high and some steps have been taken (e.g. the Dept. of Horticulture has been given evidence and a proposal to support scaling), but formal policy inclusion is just starting. PZTM is not yet in the state’s standard “recommended practices” list or subsidy programs. However, Bihar’s policy environment is trending toward supporting such innovations (the state’s Agricultural Roadmap

emphasizes sustainable practices and residue management, which align with PZTM). *Use*: Low – At present, no large-scale government program is deploying PZTM; use of policy tools (like funding, schemes) to promote PZTM is minimal. Extension officers have been involved informally via the pilot, but it's not an official part of their job targets statewide. *Bottlenecks*: The main bottleneck is the *time and process* to institutionalize a new practice. Government budgeting and program cycles mean PZTM support might only roll out state-wide after proving in pilot districts and getting administrative approvals. Another bottleneck could be the inertia of the system – extension staff need training and motivation to switch from conventional advice (ridging, etc.) to advising zero-till methods. If not directed from the top, they may stick to familiar recommendations. *Scaling Strategies*: The strategy is to actively engage policy champions and integrate PZTM stepwise. In the near term, the project seeks to **mainstream PZTM into existing schemes**: for instance, leveraging the National Mission on Sustainable Agriculture or RKVY funds to cover PZTM demonstrations and inputs in target districts. The Dept. of Agriculture is expected to issue advisories to all district officers to promote PZTM and include it in seasonal extension plans. Over five years, a goal is to incorporate PZTM into the *State Agricultural University curriculum and extension training modules* so new agri staff learn it by default. Another tactic is formal MoUs or joint implementation plans between CIP and the government, so that after the project, the state continues the scaling momentum. By Year 5, the vision is that the practice is “institutionalized” – meaning PZTM is part of the government's own program (with or without CIP/GIZ). Achieving that will involve demonstrating quick wins (so policymakers see the value), continuous lobbying with evidence, and possibly linking PZTM to flagship initiatives (for example, Bihar could claim climate action or women's empowerment gains from rolling it out). If successful, the *readiness* of this governance innovation will be marked by official programs in place, and *use* will be evident as public extension routinely carries PZTM forward. This complement is less tangible than others, but it is critical for sustainability: without embedding in institutions, scaling might stall once external project support ends.

5. Related Innovations – Seed Systems and Pest Management: *Innovation*: Two additional complementary areas are worth noting – **quality seed potato supply** and **integrated pest management tailored to PZTM**. While not “innovations” in the sense of new gadgets, improving these systems complements PZTM's scaling. *Readiness*: For seed systems – moderate. Bihar has some seed production programs and private suppliers, but small farmers often recycle seed tubers. CIP's outreach to companies like PepsiCo hints at linking PZTM growers to certified seed or even using PZTM plots to produce seed potatoes. For pest/disease management – moderate. Generally, PZTM faces fewer weeds and potentially less late blight risk (due to microclimate differences), but new dynamics like mulch-attracted pests require adapted IPM strategies. CIP has the expertise, and initial guidance (e.g. monitoring if early blight occurs under mulch) is being incorporated. *Use*: Current use of improved seed among pilot farmers is limited (some demos provided high-quality seed, but many still used farm-saved). Use of pest management specific to PZTM is low – farmers mostly applied their usual practices, though mulch suppressed some weeds/pests naturally. *Bottlenecks*: If seed quality isn't improved, PZTM's yield advantage could be undercut – farmers might blame the method for poor yields that are actually due to degenerated seed. As for pests, rodent control emerged as a significant bottleneck, as discussed, and there's also a need to watch for any new disease patterns (e.g. tubers on soil surface might face more rodent or insect exposure). *Scaling Strategies*: Work on **seed systems** by aligning with government seed

programs or private sector. For example, encourage that mini-tubers or foundation seed of popular varieties (Kufri Jyoti, Kufri Pukhraj, etc.) are made available in PZTM clusters, so adopters start with good planting material. The interactions with PepsiCo/McCain could evolve into contract farming where companies provide quality seed to farmers and then buy back the produce – this would ensure a virtuous cycle of good seed and market. On **pest management**, the scaling plan includes training extension officers in PZTM-specific troubleshooting. Modules will cover rodent prevention (field sanitation, community trapping campaigns, possibly using netting on plot boundaries). Also, extension will advise on any needed tweaks in fungicide schedules – though initial results show no major disease uptick, vigilance is planned. By compiling and disseminating these complementary agronomic practices (as a “PZTM package”), both *readiness* and *use* of these support innovations will improve. Essentially, PZTM will be promoted not in isolation but as part of an **innovation bundle** (good seed + zero till + mulch + pest control), thereby addressing bottlenecks that could otherwise constrain yield or farmer acceptance.

In summary, **Scaling Readiness analysis** indicates that while the core PZTM technology is mature (high readiness) and effective, several complementary innovations are at varying stages and must be advanced in parallel. Key bottlenecks like mulch supply and mechanization currently lag in readiness and use, posing risks to scaling. The proposed strategies – from enhancing extension methods and policy support to technical fixes for mulch and machinery – are designed to **raise the readiness and adoption of these complementary solutions**. By doing so, the overall innovation package (core + complements) becomes robust enough to reach >50,000 farmers. This approach follows CGIAR’s Scaling Readiness methodology: strengthen or substitute low-readiness elements and exploit high-readiness ones to improve the package’s “**Innovation Use Score**”. Through iterative monitoring, the scaling team will identify which bottlenecks are most limiting adoption and then implement targeted actions (e.g. if straw availability remains a problem in Year 2, invest more in that area). The end goal is a coherent set of innovations all performing at high readiness and uptake – at that point, PZTM can truly scale with minimal external support. This strategic package perspective increases the likelihood of sustainable, widespread impact rather than isolated pockets of success.

Appendix C - BMC-lite Analysis for the Bihar PZTM Delivery Model

Using a Business Model Canvas lens, we outline the core elements of the PZTM scaling model tailored to Bihar:

- **Value Proposition:** PZTM offers smallholder farmers a *win-win* value: **higher profitability, lower inputs, and climate resilience**. By eliminating plowing and utilizing on-farm straw, farmers save labor, fuel, and water (cultivation cost drops ~25–30% per ha), while often maintaining or increasing yield (+10–15% on average, with cases of up to 20–30% higher yields in late-planted or problematic soils). This translates to improved net income – for example, pilot farmers saw profits per hectare roughly double compared to conventional practice. PZTM also confers resilience benefits: the straw mulch retains moisture and cools the soil, allowing potatoes to avoid late-season heat and water stress. In essence, farmers get a more productive crop with fewer resources and less risk. Another key value is **labor efficiency and inclusion**. PZTM drastically reduces hard labor tasks (no plowing, no ridge making, minimal weeding and easier harvest). This not only lowers cost but also reduces drudgery – a boon in a context of labor shortages and increasing wage rates. Women in particular benefit, as they can manage PZTM plots with less reliance on male labor or hired tractors. The method thereby empowers women

farmers and saves time that can be used for other activities. For the broader system (government and society), PZTM's value proposition includes **environmental services** – it curbs open burning of rice straw (cutting air pollution), improves soil health via organic matter addition, and lowers greenhouse gas emissions by ~40%. This makes it a climate-smart innovation that aligns with public goals. In summary, the value proposition can be summed up as: *“Grow more potatoes at less cost and effort, while improving your soil and environment.”* Farmers realize immediate economic gains (better returns on investment) and longer-term soil fertility benefits, all of which make PZTM an attractive proposition in Bihar's rice-potato systems.

- Customer Segments:** The primary customers (end-users) are **smallholder potato farmers in Bihar**, especially those cultivating potatoes after rice. Bihar has ~800,000–900,000 such farmers, largely with small plots (average ~0.4 ha under potato). The scaling effort initially focuses on existing potato-growing districts – e.g. Nalanda, Vaishali, Patna, Rohtas – where potato area is high and farmers are already invested in the crop. Within this broad segment, particular sub-segments are prioritized: **women farmers** (often organized in SHGs via JEEViKA) are a key target, both because they have a major role in potato cultivation and because PZTM can enhance their agency by reducing heavy labor needs. The aim is for at least 50% of adopters to be women and to have 1,000+ women in leadership roles as community promoters. Another sub-segment is **resource-poor and marginal farmers** – those with limited land and capital who stand to benefit from a low-cost innovation. PZTM is well-suited for marginal farmers (even those with as little as 0.1 ha or in waterlogged/sodic patches that they couldn't cultivate before). The technology's scalability also means it can serve **medium/large progressive farmers** looking to optimize input use and adopt climate-friendly practices, but the core push is with smallholders who need the cost savings most. Secondary customer segments include stakeholders who are “customers” in a broader sense: for example, the **Government of Bihar** (Department of Agriculture/Horticulture) is a customer of the solution in that it helps them achieve policy goals (higher farm incomes, sustainable agriculture) – they “buy into” PZTM by integrating it into programs. Similarly, value-chain actors like **potato processing companies** or **seed suppliers** are customers in that they benefit from the outcome (more reliable potato production, new sourcing opportunities) and may invest resources to support scaling. But the primary segmentation is geographic (potato districts) and demographic (smallholders, with a focus on women and poorer households). Lastly, **delivery partners** themselves form a segment: extension agents, NGO field staff, and community volunteers who will be *users of the approach* to reach farmers. They need to be convinced of PZTM's value (so they promote it enthusiastically) – thus, we treat them as an important customer segment for training and motivation. Their needs (training materials, evidence) are being addressed so they can effectively “sell” the innovation to farmers.
- Delivery Channels:** The PZTM scaling model employs multiple complementary channels to reach farmers with both knowledge and initial support. **Community-based demonstration and learning** is a core channel: in each target village, lead farmers or SHGs will host demo plots (often on a portion of land to compare PZTM vs traditional). Field days and cross-visits allow neighboring farmers to observe results first-hand – this direct exposure was critical in the pilot to convert skeptics into adopters. **Video-mediated extension** is another channel: Digital Green's local-language videos on PZTM are screened in group meetings (SHG meetings, village gatherings). Battery-powered pico

projectors and facilitator-led discussions bring the practice to life with visual instruction. This channel is powerful for reaching semi-literate audiences at scale and ensuring consistency of message. **Government extension services** form a third channel: the state's network of Block Agriculture Officers, village extension workers (Krishi Salahkars), and Subject Matter Specialists will incorporate PZTM in their regular farmer outreach. They will identify interested farmers, provide technical advice, and include PZTM topics in agri-training camps. Because they are a trusted source, their endorsement via trainings and advisories lends credibility. **JEEViKA's SHG network** is a critical grassroots channel: JEEViKA mobilizers and community resource persons will introduce PZTM in group meetings, facilitate collective action (like organizing straw collection), and follow up with women farmers individually. The SHG structure, with regular meetings and savings-credit activities, provides a ready platform to disseminate information quickly across tens of thousands of members. **Printed and digital materials** also support delivery: CIP has created handouts, agronomy bulletins, and will share these through agri-fairs and input dealers. WhatsApp groups for field staff and progressive farmers will be used for quick problem-solving tips (leveraging increasing smartphone use in rural areas). Lastly, **media and events** act as supplementary channels – local radio might broadcast PZTM success stories, and district-level “Kisan Melas” (farmer fairs) will feature PZTM exhibitions. By combining high-touch channels (demos, extension agent visits) with low-touch but wide-reach channels (video, radio), the project ensures both depth of understanding and breadth of awareness. Importantly, these channels are coordinated: for example, an SHG screening of a video is typically followed by scheduling a hands-on demo in that community, creating a reinforcement loop. The use of existing social infrastructure (SHGs) and government networks drastically lowers the cost of delivery per farmer while enabling scale. In summary, the delivery strategy is **multi-channel** – leveraging community learning, digital content, public extension, and women's networks – to efficiently and effectively reach the target of 50,000+ farmers.

- **Cost & Revenue** (Financial Logic): The PZTM scaling model is driven by cost-efficiency and sustainability rather than direct revenue, as it's a public-good oriented intervention. **Cost structure:** On the implementation side, major cost components include training and capacity building (developing content, training thousands of farmers and facilitators), demonstrations (providing some seed, straw, or inputs for demo plots), coordination and monitoring (staff salaries, travel, data collection), and equipment (e.g. projectors for video dissemination, any pilot machinery for planting). The ongoing GIZ-supported project provides much of the funding for these through 2025, and a detailed budget has been outlined covering training, dissemination, M&E, and technical support from CIP. Thereafter, integration into government programs will shift some costs to the public extension budget (e.g. extension officers' time is salaried by Govt, some demonstration costs can be covered by schemes like RKVY). At the farmer level, the **cost structure** of adopting PZTM is favorable: farmers *avoid* the expense of land preparation (plowing, ridge making), which in Bihar can be ₹5,000–₹8,000 per acre (tractor hire, diesel) – these costs drop to near-zero under PZTM. They also reduce spending on irrigation (50% fewer irrigations) and weeding/herbicides. The main added “cost” is the labor or effort to spread mulch, but this is offset by elimination of digging at harvest (no heavy lifting; potatoes are easily picked from under straw). In practice, pilots showed total labor days per ha dropped from ~197 to 137. Thus, for farmers, PZTM *reduces* cost of cultivation by roughly ₹30,000/ha (US\$400), a significant savings. **Revenue streams:** The project itself doesn't generate commercial revenue (it's not selling a product to farmers), but it creates value

that can be monetized or lead to financial flows indirectly. For farmers, the revenue is increased crop income – PZTM farmers often get equal or higher yields than conventional, and even when yield is equal, their profit is higher due to cost savings. For example, one case showed productivity 32.5 → 36.2 t/ha and net income jumping from ₹79.5k to ₹154.5k/ha under PZTM. This improved income is essentially the “customer revenue” from adopting the innovation. Another potential revenue logic is through **carbon credits or climate finance**: since PZTM reduces emissions (0.72 tons CO₂/ha avoided from burning, plus soil carbon gains), there is an opportunity to quantify this and attract payments for mitigation. At an estimated \$21.6/ha carbon benefit, if 50,000 ha eventually adopt, that’s over \$1 million worth of CO₂ abatement – not currently realized as revenue, but a figure that could be used to lobby for climate funding or carbon credit projects that benefit farmers or the program. For scalability, **financial sustainability** is key: the model is to front-load external and public investment to build momentum, after which the practice is privately profitable enough that farmers continue it on their own (no ongoing subsidy needed for PZTM itself). In fact, farmers saving money is itself the mechanism that sustains adoption. The role of the government and partners is to absorb initial learning and promotion costs. As scale is achieved, unit costs per farmer drop (economies of scale in training via videos, etc.). If in later years a private sector partner like a food company comes in, they might contribute funds (e.g. to train their contract farmers in PZTM) – that would be an added revenue/investment stream to expand reach. In conclusion, the financial logic rests on **making farming cheaper and more profitable for producers** (driving spontaneous uptake), using **smart subsidies/investments initially** (for training, demos, perhaps machinery) but designing those to taper off as the innovation becomes self-propelling. By embedding PZTM into normal extension and farmers’ own cost calculations, the model avoids long-term dependency on external funding, which is crucial for scaling beyond the project’s life.

- **Key Partnerships:** The scaling model is inherently partnership-based, leveraging the comparative advantages of various actors. **Government of Bihar (Dept. of Agriculture & Dept. of Horticulture):** a pivotal partner providing institutional backing, outreach infrastructure, and legitimacy. They chair the coordination committee and integrate PZTM with state programs, ensuring that scaling efforts align with public initiatives (e.g. climate-resilient agriculture, zero-burn campaigns). Their extensive field staff network is a delivery arm for the innovation. **Bihar Rural Livelihoods Promotion Society (JEEViKA):** brings the vast network of women’s SHGs into play. As a partner, JEEViKA mobilizes communities, hosts training via its community resource persons, and helps target women and marginal groups effectively. They essentially provide a social capital network that increases trust and adoption at grassroots. **Digital Green:** an expert partner for ICT-enabled extension – they produce high-quality training videos and build capacity of local mediators to use them. DG’s partnership is key for scaling efficiently; their existing experience in Bihar (with NRLM and others) means they can deploy digital extension innovations quickly. **International Potato Center (CIP):** the technical backbone partner – CIP introduced PZTM, has agronomists and scaling specialists guiding the approach, and will continue R&D (e.g. troubleshooting, varietal interactions, mechanization trials) to refine the innovation. CIP also ensures monitoring and evaluation rigor and provides global knowledge (linking experiences from Assam, Bangladesh, etc., to Bihar). **GIZ (and the BMZ i4Ag project):** the catalytic funding and advisory partner – GIZ not only co-funds the scaling efforts but also contributes expertise in scaling strategies (bringing tools like the Scaling Scan, and facilitating cross-learning with similar projects). GIZ’s convening

power helps engage policy makers and align the project with donor priorities on climate and innovation. **Local Research and Training Institutes:** e.g. ICAR's Central Potato Research Station (CPRS) in Patna, or Bihar Agricultural University. These partners can validate results, adapt recommendations to local conditions, and help with capacity building (training extension officers, etc.). They also lend credibility among scientific and policy circles in India. **Private Sector Partners:** though in early stages, entities like PepsiCo and McCain (potato processors) are engaged in dialogue. They could become key partners by integrating PZTM in their supply chains – for instance, providing market assurance to PZTM adopters or investing in scaling (if they see it secures raw material or reduces their Scope 3 emissions). Similarly, agri-tech companies or equipment manufacturers could partner to develop and supply machinery or inputs (e.g. a local manufacturer partnering to produce zero-till planters, or a startup offering digital advisory services tailored to PZTM farmers). **Community Institutions:** besides SHGs, village panchayats (local governments) and farmer clubs can partner at micro-level to support things like communal rodent control drives or managing straw resources collectively. These partnerships are less formal but important for local acceptance. **Financial partners:** if microfinance institutions or banks provide credit lines for things like purchasing seed or machinery for PZTM farmers, they become part of the partnership ecosystem (ensuring farmers have means to invest in the new practice). In summary, the PZTM business model is a multi-actor coalition. Each partner has clear roles: government provides scale and sustainability, NGOs provide innovation in delivery, CGIAR (CIP) provides science and convening, donors provide risk capital and scaling know-how, and community organizations provide reach and trust. This network of partnerships greatly amplifies what any single actor could do – it is the crux of the scaling strategy. As evidenced in the proposal, roles and coordination mechanisms have been explicitly defined to make this collaboration work seamlessly. The long-term vision is that even after donor support ends, the core partners (govt, JEEViKA, maybe private sector) will continue championing PZTM, with CIP in an advisory role, thus sustaining the innovation through institutional partnerships.

Appendix D - GenderUp Analysis (Responsible Scaling – Gender and Social Inclusion)

Scaling PZTM responsibly in Bihar necessitates attention to gender dynamics and social inclusion, ensuring that women and marginalized groups benefit and that no one is inadvertently harmed. Below we assess risks and opportunities for key social segments and suggest safeguards/design adaptations:

- **Women (SHG Members and Female Farmers): Opportunities:** PZTM has been intentionally designed and piloted with women farmers at the forefront, offering several empowerment gains. It reduces labor drudgery (no heavy tillage or digging), which directly benefits women who bear ~60% of agricultural labor in Bihar. Women adopters report that they can manage potato cultivation largely by themselves with PZTM, instead of depending on male family members or hired plowmen. This shift gives them greater control over the farming process and decision-making. The Digital Green gender study found women valued the reduction in labor and input costs, and noted improvements in the quality and quantity of produce – validating the hypothesis that PZTM can enhance women's farming outcomes. Women-led SHGs have been crucial for disseminating the practice, and this community approach builds **women's leadership** (e.g. female Village Resource Persons acting as local extension agents). In fact, one scaling objective is to have 1,000+ women as PZTM promoters or lead farmers, strengthening their public role

in agriculture. PZTM also frees up some of men's labor (since men usually handle plowing and irrigation) and slightly increases women's on-field tasks (like arranging mulch). However, the net effect is positive: lighter tasks for women replace some previously heavy tasks done by men, potentially giving women more say and visibility in those farming operations. **Risks:** Traditional gender norms could impede women's access or agency in scaling. In the pilot, many women had to **seek permission** from husbands or family to try PZTM, as families were initially unconvinced about this new method. This indicates a risk that women early-adopters might face skepticism or resistance at home. If a woman's PZTM plot were to fail (due to factors outside her control, like extreme weather), it could reinforce negative stereotypes about women's farming decisions. Another risk is that once PZTM proves profitable, men might reassert control over the activity. It's a common pattern that men take over enterprises deemed "successful," potentially sidelining women who pioneered the practice. There's also the risk of increased **time burden** on women if not monitored – while PZTM reduces total labor, if women end up doing all tasks (including those men used to do like spraying or hauling straw), their workload could temporarily spike. However, the data suggests female labor went up only 2% while male labor dropped 33%, so it's more of a redistribution than a net increase. **Safeguards/Adaptations:** To leverage opportunities and mitigate risks, the scaling strategy has a **gender-intentional design**. It includes a PZTM Gender Strategy (developed by GIZ and CIP) to ensure women have equitable access to training, information, and inputs. Trainings are organized at times/places convenient for women, often through SHGs, and led by female VRPs to encourage participation. The project also emphasizes couple engagement – inviting husbands and male family members to key demonstrations or training sessions. This proved effective in pilots: once men saw the bumper harvests from their wives' PZTM plots, their perceptions changed from doubt to support. Such joint sensitization can reduce household-level resistance. To prevent men's takeover, the program will promote **recognition of women's success** (e.g. awards to outstanding women PZTM farmers, featuring women champions in videos), thereby validating women's expertise in the community. Moreover, linking women farmers to credit and markets (perhaps via SHG federations or FPOs) will strengthen their autonomous benefit from PZTM. The gender study recommends continued support for women's leadership and addressing any emerging gender-based constraints (for example, if handling mulch at scale is difficult, introducing small mechanized straw spreaders that women can operate). In summary, by embedding PZTM within the women's SHG platform and actively involving/educating men alongside, the scaling effort aims to deliver **empowerment outcomes (increased agency, income, confidence)** for women while minimizing patriarchal pushback. Ongoing gender-focused monitoring (feedback from women's groups) will be used to promptly adjust strategies if, say, women report new challenges or if their role is being overshadowed.

- **Landless and Laboring Groups: Opportunities:** Indirectly, PZTM could benefit landless rural people if approached thoughtfully. For instance, landless women often participate in SHGs – while they may not have their own land to implement PZTM, they could engage in collective farming (SHG members leasing a plot together to grow potatoes using PZTM, thus sharing profits). The simplified cultivation might make it feasible for groups of landless women or youth to rent small parcels and succeed in cultivation, which they might not attempt with conventional methods due to high input requirements. Additionally, landless individuals could find new roles in the PZTM ecosystem: the innovation calls for services like straw collection, guarding mulched fields from stray cattle, or custom hiring of planters.



These needs can be turned into entrepreneurial ventures for landless laborers (e.g. a group of landless youth contract to supply mulch to farmers for a fee, or provide rodent control services across a village). *Risks:* The more immediate impact of PZTM on landless agricultural laborers is likely **reduced labor demand** in potato farming, which could negatively affect those who rely on wage labor. Traditional potato cultivation employs many laborers for land prep (plowing, ridge formation, etc.) and harvesting (digging tubers). PZTM cuts those tasks; data indicates male labor days in potato dropped by one-third under PZTM, which often means fewer hiring of laborers (who in rural Bihar are frequently landless or land-poor men). Female labor days didn't drop much, but some tasks like weeding and earthing-up were eliminated, possibly reducing hiring of women laborers too. So there is a **trade-off**: what's good for adopter farmers (less labor cost) can be bad for laborers (fewer paid days). If PZTM scales widely without alternatives in place, landless families might lose a source of income, exacerbating rural inequality. Another risk is that landless households, who often rely on common resources, could be disadvantaged if, say, rice straw which might have been free for anyone to collect after harvest is now all claimed for mulch by landowners. If not managed, this could create community tensions or reduce fodder availability for those who don't farm but keep livestock. *Safeguards/Adaptations:* The responsible scaling approach should monitor labor dynamics. One mitigation is to facilitate **alternative employment or income for laborers** in the same innovation sphere. For example, as mentioned, training interested landless youth or women to become PZTM service providers – operating a rented planter, running a straw baling operation, or being community facilitators paid to support multiple farmers – can create new skilled roles. The project can coordinate with JEEViKA to identify landless SHG members who want to participate economically (perhaps providing them with skills or micro-loans to start enterprises around PZTM, like vermicomposting with leftover straw, or small trading of seed potato). Additionally, promoting **collective farming models** (SHG groups pooling land or contracting with landowners for a share of crop) could enable landless women to directly benefit from PZTM yields and profits. The extension messages also emphasize that freed-up labor (especially male labor) can be redirected to other productive activities, meaning households can diversify income – the program could link laborers to training for off-season or off-farm work (in collaboration with rural development schemes). To address the straw resource issue, a community management approach is needed: ensure that using straw for mulch doesn't deprive those who need it for fodder. This might involve encouraging farmers who adopt PZTM (and hence don't burn straw) to share or sell surplus straw to neighbors with livestock at a fair price, rather than conflict. The coordination with village leaders and JEEViKA could help negotiate such arrangements. In essence, while some reduction in menial farm labor is expected (and indeed is part of progress), the scaling initiative should **track socio-economic impacts on laborers** and support measures that help landless groups transition – whether into better-paying skilled roles in the value chain or into other livelihoods – so they are not left worse off. This aligns with a do-no-harm ethos and ensures that the gains of PZTM are inclusive.

- **Caste Dynamics (Social Inclusion):** *Opportunities:* Bihar's rural populace includes historically disadvantaged castes (Scheduled Castes, OBCs, etc.), many of whom are small/marginal farmers or landless. PZTM's low-cost nature can be especially beneficial for poorer farmers who often belong to lower castes; it lowers barriers to entry in potato cultivation (you don't need to hire a tractor or buy lots of fertilizer upfront). The pilot and gender research intentionally covered diverse social groups – OBC, SC, and general caste

farmers were all represented among participants. Because dissemination is through SHGs and community groups which typically cut across minor caste lines (or at least organize the marginalized as a collective voice), the approach can elevate lower-caste women as innovators in their communities. If scaled well, PZTM could serve as a case where an agricultural innovation *reduces inequality*: marginal farmers (often lower caste) improving their production and income without needing large land or capital. *Risks*: Social hierarchy and caste dynamics could affect equitable adoption. Higher caste or wealthier farmers might initially dismiss PZTM (since it was piloted primarily with women and poorer households as an “easy, low-input” method) – but if it succeeds, they might later dominate its benefits, potentially overshadowing the smaller farmers. There is also a risk that extension efforts might inadvertently favor villages or groups that are more accessible or vocal, which can correlate with higher-caste communities. Marginalized communities (e.g. SC hamlets) might get less attention unless explicitly targeted. Another concern is decision-making power: in some villages, if local elites or dominant caste leaders are not involved or convinced, they could discourage others from trying the innovation (“if it were any good, we would be doing it”). Caste-based land ownership patterns (many lower-caste households have very tiny or no land) mean that without special effort, the landless or near-landless (often SCs) could be left out of direct adoption. *Safeguards/Adaptations*: The scaling strategy can promote **social inclusion** by leveraging JEEViKA’s mandate to reach the poorest. JEEViKA SHGs often are formed among the most vulnerable women, including many from SC or ST communities, ensuring that outreach inherently covers those groups. Extension micro-plans will be made for each panchayat, with an eye to covering all habitations – including Dalit tolas (hamlets) – not just main village clusters. One adaptation is to intentionally set up demonstration plots in fields owned by lower-caste farmers or in areas accessible to them, thereby sending a message that this innovation is for everyone. The project’s monitoring will disaggregate adoption data by gender and social category (SC/ST/OBC) to track who is benefiting. If any group is lagging, corrective actions (like focused training camps for those communities or additional support) will be taken. Moreover, care will be taken in group settings to ensure inclusive participation – e.g. if an SHG meeting has mixed caste presence and some women are hesitant to speak due to social norms, facilitators will use methods to draw out everyone’s views (perhaps separate focus group discussions for different social segments if needed to ensure open feedback). Caste dynamics around straw (for instance, if traditionally lower caste people could collect straw after harvest from upper caste landowners’ fields for fodder) should be considered – the program might encourage that mulching one’s field and then giving the used straw (post-harvest) to cattle owners is an option, thus maintaining a form of sharing. Lastly, involving local governance (Panchayat heads, who in many cases now include members of SC due to reservation policies) in the coordination committees at block or district level can help voice the needs of marginalized groups. By conscientiously implementing these measures, the project aims for **broad-based inclusion** such that PZTM does not become an innovation only for the “privileged” but truly reaches the poorer and socially disadvantaged farmers who can benefit greatly from its low-cost, high-return nature.

- **Youth and Next-Generation Farmers:** *Opportunities*: Engaging youth in PZTM scaling can yield multiple benefits. Rural youth often bring energy, openness to new ideas, and familiarity with technology – traits that fit well with PZTM’s promotion via digital tools and new practices. Younger farmers or agri-entrepreneurs could become **local champions** of PZTM, helping change mindsets in communities that might be set in older ways. For

example, a tech-savvy youth might operate the pico projector for video showings, or manage a WhatsApp group to disseminate PZTM tips. There are also entrepreneurship opportunities for youth around PZTM: custom hiring services for mechanized planting or straw baling, as mentioned, could attract unemployed rural youth if they see a viable business. Additionally, PZTM's labor-saving aspect may appeal to youth by reducing the drudgery that often drives them away from farming. With PZTM, farming potatoes could be less back-breaking and more "modern," potentially retaining or bringing back some young people to agriculture. *Risks:* One risk is that many rural youth in Bihar are migrating or seeking non-farm jobs; they may be disengaged from agriculture and thus not initially interested in adopting or promoting PZTM. If not involved, the scaling relies solely on older farmers, who might be more conservative or physically less able to experiment. Another risk is that if mechanization is introduced without youth involvement, it could be dominated by already established older businesses (e.g. existing tractor owners) – missing the chance to empower youth. If scaling doesn't address youth explicitly, a gap could form where older farmers adopt but the next generation doesn't learn or continue the practice, threatening sustainability. *Safeguards/Adaptations:* The project can incorporate **youth engagement strategies**. For instance, recruit and train rural youth as "PZTM facilitators" or community technicians. In each block, young graduates (perhaps those involved in agri programs or returnee migrants) could be enlisted to support data collection (using tablets/phones for M&E) and troubleshooting. This gives them a stake and some earning, while improving project implementation. The mention of exploring a PhD student's involvement in adoption monitoring shows an openness to involving young researchers, which could extend to local youth participation in citizen science (e.g. youth helping measure demonstration plot outcomes). To increase youth interest, PZTM can be framed as a **modern, climate-smart farming technique** – connecting with youth's aspirations for innovation and environmental stewardship. Showcasing PZTM at colleges or vocational training centers (like ITIs or ag polytechnics) could integrate it into youth education. On the entrepreneurship side, the initiative could coordinate with schemes like Start-up Village Entrepreneurship Program (SVEP) or other livelihood programs to provide small grants or loans to youth who want to start services (for straw aggregation, machinery, or input supply for PZTM communities). By highlighting successful young role models (say, a young farmer who adopted PZTM and improved his family's income, or a group of youth managing a custom planting service), the project can inspire peers. In summary, while youth are not the traditional potato farmers in many villages, making them part of the scaling process ensures **intergenerational transfer of knowledge** and injects dynamism into the initiative. It prevents the risk of aging farmer populations being the only adopters. Engaging youth also aligns with Bihar's focus (and CGIAR's emphasis) on attracting youth to agriculture. Therefore, the scaling design will include youth-tailored communication (perhaps using social media or youth networks) and opportunities for them to benefit economically, turning what could be a risk of exclusion into an opportunity for renewal in the farming community.